Introduction to Engineering Labs

Alveary

Alveary

Alt. Science Options

Introduction to Engineering Labs

GRADE(S): 9-12

About the Course

Note that labs are an essential part of science in which students engage with the Things they are reading about and practice the scientific method.

This topic is included in the following course(s): Science: Introduction to Engineering

Scheduling

GRADE	SCHEDULE INFO.	BOOKS
9-12	Introduction to Engineering Labs 1 time/week 60 min	

Sample Weekly View

Day 1	Day 2	Day 3	Day 4	Day 5
Science: Introduction to Engineering				
Introduction to Engineering Lessons Nature Walks & Scouting: Grades 9-12	Introduction to Engineering Lessons	Introduction to Engineering Lessons	Introduction to Engineering Lessons Nature Notebook: Grades 9-12	Introduction to Engineering Lessons Introduction to Engineering Labs

Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ View Basic Supplies

(No Subject Supplies Assigned)

Quick Links

(No Quick Links Assigned)

Click THIS text or scan the QR code for links.

Introduction to Engineering Labs

How To Approach

Introduce

• Regardless of how many days are required to complete a particular activity, every Science Lab has the same flow, which follows the scientific method.

• Relevant concept(s) are introduced in the text.

• Your notebook entry begins with the introductory/prelab narration, including relevant information that you have read or previously experienced, what you plan to do in the lab, and any hypothesis or anticipated result.

Lab Procedure

• Perform the procedure according to the instruction, recording in your notebook what you do as it happens. This can be a challenge, but is an extremely important skill.

• Record all data and observations in the lab notebook.

Analysis & Conclusions

• After all data is collected, analyze the results by considering how the data reflects the introduced concepts and whether the hypothesis is supported by the data.

• If the data and observations do not support the hypothesis, reflect on why and what further testing would be interesting or helpful.

Introduction to Engineering Labs

[Click THIS text or scan the QR code for links.]

Term 1

WEEK 1

⬚ 60m Introduction to Engineering Labs - Lesson 1

Materials: internet access

→ PROJECT
If there is a design problem in your home or community that you would like to tackle, use this time to research, design, and execute your solution.

If you'd like to sample different areas of engineering, begin with the structural engineering project linked, The Leaning Tower of Pasta. Or you might find your own idea by exploring the Science Buddies website. The filters on the left side of the page allow you to narrow your results. Once you have chosen a project, let your teacher know if you need any materials. Your project might take the whole year or you might finish it in just a few weeks. Whenever you finish your project, select another one! Record your work and your observations in your lab notebook.
∞ Website Link: The Leaning Tower of Pasta
∞ Website Link: Science Buddies

WEEK 2

⬚ 60m Introduction to Engineering Labs - Lesson 2

Materials: by student choice & notebook

→ PROJECT
Continue working on your project.

WEEK 3

⬚ 60m Introduction to Engineering Labs - Lesson 3

Materials: by student choice & notebook

→ PROJECT
Continue working on your project.

WEEK 4

⬚ 60m Introduction to Engineering Labs - Lesson 4

Materials: by student choice & notebook

→ PROJECT
Continue working on your project

WEEK 5

⬚ 60m Introduction to Engineering Labs - Lesson 5

Materials: by student choice & notebook

→ PROJECT
Continue working on your project.

WEEK 6

⬚ 60m Introduction to Engineering Labs - Lesson 6

Materials: by student choice & notebook

→ PROJECT
If there is a design problem in your home or community that you would like to tackle, use this time to research, design, and execute your solution.

If you'd like to sample different areas of engineering, try the materials engineering project linked, Pounding Papyrus. Or you might find your own idea by exploring the Science Buddies website. The filters on the left side of the page allow you to narrow your results. Once you have chosen a project, let your teacher know if you need any materials. Your project might take the whole year or you might finish it in just a few

Introduction to Engineering Labs

[Click THIS text or scan the QR code for links.]

Term 1

weeks. Whenever you finish your project, select another one! Record your work and your observations in your lab notebook.
∞ Website Link: Pounding Papyrus
∞ Website Link: Science Buddies

WEEK 7
⬚ **60m Introduction to Engineering Labs - Lesson 7**

> Materials: by student choice & notebook

→ PROJECT
Continue working on your project.

WEEK 8
⬚ **60m Introduction to Engineering Labs - Lesson 8**

> Materials: by student choice & notebook

→ PROJECT
Continue working on your project.

WEEK 9
⬚ **60m Introduction to Engineering Labs - Lesson 9**

> Materials: by student choice & notebook

→ PROJECT
Continue working on your project.

WEEK 10
⬚ **60m Introduction to Engineering Labs - Lesson 10**

> Materials: by student choice & notebook

→ PROJECT
Continue working on your project.

WEEK 11
⬚ **60m Introduction to Engineering Labs - Lesson 11**

> Materials: by student choice & notebook

→ PROJECT
Continue working on your project.

WEEK 12
⬚ **60m Introduction to Engineering Labs - Lesson 12**
Exam Week

→ EXAMS
Answer exam questions from:
Project Work

Introduction to Engineering Labs

Click THIS text or scan the QR code for links.

Term 2

WEEK 1

60m Introduction to Engineering Labs - Lesson 13

Materials: by student choice & notebook

→ PROJECT
If there is a design problem in your home or community that you would like to tackle, use this time to research, design, and execute your solution.

If you'd like to sample different areas of engineering, try the mechanical engineering project linked, Choosing the Best Gear Ratio for Maximum Bike Speed. Or you might find your own idea by exploring the Science Buddies website. The filters on the left side of the page allow you to narrow your results. Once you have chosen a project, let your teacher know if you need any materials. Your project might take the whole year or you might finish it in just a few weeks. Whenever you finish your project, select another one! Record your work and your observations in your lab notebook.
∞ Website Link: Choosing the Best Gear Ratio for Maximum Bike Speed
∞ Website Link: Science Buddies

WEEK 2

60m Introduction to Engineering Labs - Lesson 14

Materials: by student choice & notebook

→ PROJECT
Continue working on your project.

WEEK 3

60m Introduction to Engineering Labs - Lesson 15

Materials: by student choice & notebook

→ PROJECT
Continue working on your project.

WEEK 4

60m Introduction to Engineering Labs - Lesson 16

Materials: by student choice & notebook

→ PROJECT
Continue working on your project.

WEEK 5

60m Introduction to Engineering Labs - Lesson 17

Materials: by student choice & notebook

→ PROJECT
Continue working on your project.

WEEK 6

60m Introduction to Engineering Labs - Lesson 18

Materials: by student choice & notebook

→ PROJECT
Continue working on your project.

WEEK 7

60m Introduction to Engineering Labs - Lesson 19

Materials: by student choice & notebook

→ PROJECT
If there is a design problem in your home or community that you would like to tackle,

Term 2

use this time to research, design, and execute your solution.

If you'd like to sample different areas of engineering, try the electronics engineering project linked, Make Your Own Low-Power AM Radio Transmitter. Or you might find your own idea by exploring the Science Buddies website. The filters on the left side of the page allow you to narrow your results. Once you have chosen a project, let your teacher know if you need any materials. Your project might take the whole year or you might finish it in just a few weeks. Whenever you finish your project, select another one! Record your work and your observations in your lab notebook.
∞ Website Link: Make Your Own Low-Power AM Radio Transmitter
∞ Website Link: Science Buddies

WEEK 8

60m Introduction to Engineering Labs - Lesson 20

Materials: by student choice & notebook

→ PROJECT
Continue working on your project.

WEEK 9

60m Introduction to Engineering Labs - Lesson 21

Materials: by student choice & notebook

→ PROJECT
Continue working on your project.

WEEK 10

60m Introduction to Engineering Labs - Lesson 22

Materials: by student choice & notebook

→ PROJECT
Continue working on your project.

WEEK 11

60m Introduction to Engineering Labs - Lesson 23

Materials: by student choice & notebook

→ PROJECT
Continue working on your project.

WEEK 12

60m Introduction to Engineering Labs - Lesson 24
Exam Week

→ EXAMS
Answer exam questions from:
Project Work

Introduction to Engineering Labs

Click THIS text or scan the QR code for links.

Term 3

WEEK 1

☐ **60m Introduction to Engineering Labs - Lesson 25**

Materials: by student choice & notebook

→ PROJECT
If there is a design problem in your home or community that you would like to tackle, use this time to research, design, and execute your solution.

Or you might choose between the linked materials (Living Loom), electronic engineering projects(Obstacle-Detecting Cane with Arduino), or another project from the Science Buddies website. Once you have chosen a project, let your teacher know if you need any materials.
∞ Website Link: Living Loom
∞ Website Link: Obstacle-Detecting Cane with Arduino
∞ Website Link: Science Buddies

WEEK 2

☐ **60m Introduction to Engineering Labs - Lesson 26**

Materials: by student choice & notebook

→ PROJECT
Continue working on your project.

WEEK 3

☐ **60m Introduction to Engineering Labs - Lesson 27**

Materials: by student choice & notebook

→ PROJECT
Continue working on your project.

WEEK 4

☐ **60m Introduction to Engineering Labs - Lesson 28**

Materials: by student choice & notebook

→ PROJECT
Continue working on your project.

WEEK 5

☐ **60m Introduction to Engineering Labs - Lesson 29**

Materials: by student choice & notebook

→ PROJECT
Continue working on your project.

WEEK 6

☐ **60m Introduction to Engineering Labs - Lesson 30**

Materials: by student choice & notebook

→ PROJECT
If there is a design problem in your home or community that you would like to tackle, use this time to research, design, and execute your solution.

Or you might choose between the linked chemical formulation (Design Your Own Sports Drink), electronics engineering (Build a Magnetic Pet Door with Arduino) projects, or another project from the Science Buddies website. Note that the chemistry project only becomes an engineering project by extending it into the design phase! Once you have chosen a project, let your teacher know if you need any materials.

Introduction to Engineering Labs

Click THIS text or scan the QR code for links.

Term 3

∞ Website Link: Design Your Own Sports Drink
∞ Website Link: Build a Magnetic Pet Door with Arduino
∞ Website Link: Science Buddies

WEEK 7

⬚ 60m Introduction to Engineering Labs - Lesson 31

Materials: by student choice & notebook

➜ PROJECT
Continue working on your project.

WEEK 8

⬚ 60m Introduction to Engineering Labs - Lesson 32

Materials: by student choice & notebook

➜ PROJECT
Continue working on your project.

WEEK 9

⬚ 60m Introduction to Engineering Labs - Lesson 33

Materials: by student choice & notebook

➜ PROJECT
Continue working on your project.

WEEK 10

⬚ 60m Introduction to Engineering Labs - Lesson 34

Materials: by student choice & notebook

➜ PROJECT
Continue working on your project.

WEEK 11

⬚ 60m Introduction to Engineering Labs - Lesson 35

Materials: by student choice & notebook

➜ PROJECT
Continue working on your project.

WEEK 12

⬚ 60m Introduction to Engineering Labs - Lesson 36

Exam Week

➜ EXAMS
Answer exam questions from:
Project Work